

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design and Develop a Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern webpage using HTML5 for structure and CSS3 for styling, including navigation, content sections, service cards, and a contact form.

3. OBJECTIVE

- To use HTML5 elements to structure a modern webpage.
 - To apply CSS3 properties for page styling and layout.
 - To create a header and navigation bar.
 - To organize content into About, Services, and Contact sections.
 - To use Flexbox to arrange service articles horizontally.
 - To style form controls and a submit button.
 - To include a responsive CSS media query for smaller screens.
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4. THEORY

HTML5 provides the structural foundation of the webpage through elements such as header, nav, main, section, article, and footer. The document also uses the viewport meta tag and links a stylesheet using the link element.

CSS3 is used to control the visual presentation of the webpage. The Week 01 stylesheet contains rules for colors, typography, spacing, navigation, section cards, Flexbox layout, form controls, buttons, hover effects, and a media query that changes the service container and navigation links for screens up to 700 pixels wide.

Repository note: contact.html explicitly references style.css. The Week 01 repository currently lists contact.html inside the second program folder, while the style.css file is present in the first Week 01 program folder. The uploaded Week 01 HTML5 & CSS3 screenshot shows the styled result.

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document named contact.html.
 2. Add the character encoding and viewport meta tags.
 3. Set the page title and link the CSS stylesheet.
 4. Create a header with the webpage title and subtitle.
 5. Create navigation links for Home, About, Services, and Contact.
 6. Create the About section describing HTML5 and CSS3.
 7. Create the Services section containing three article cards.
 8. Use a container class for the service layout.
 9. Create the Contact section with name, email, message, and submit controls.
 10. Create a footer with the copyright text.
 11. Apply the CSS rules for layout, spacing, colors, forms, buttons, hover effects, and responsive behavior.
 12. Open the webpage in a browser and verify the rendered output.
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6. PROGRAM

contact.html

```
1 <!DOCTYPE html> 2
<html lang="en">
3
4 <head>
5 <meta charset="UTF-8"> <meta name="viewport"
6 content="width=device-width, initial-scale=1.0">
7
8 <title>HTML5 and CSS3 Webpage</title>
9
10 <link rel="stylesheet" href="style.css">
11 </head>
12
```

```
13 <body>
14
15     <header>
16         <h1>HTML5 & CSS3 Webpage</h1>
17         <p>Learning Modern Web Development</p>
18     </header>
19     <nav>
20         <a href="index.html">Home</a> <a
21         href="#about">About</a> <a
22         href="#services">Services</a> <a
23         href="#contact">Contact</a>
24     </nav>
25
26     <main>
27
28         <section id="about">
29             <h2>About HTML5 & CSS3</h2>
30
31             <p>
32                 HTML5 is used to create the structure of modern
33                 webpages, while CSS3 is used to design and style
34                 webpages.
35             </p>
36         </section>
37
38         <section id="services">
39             <h2>Our Services</h2>
40
41             <div class="container">
42
43                 <article>
44                     <h3>Web Design</h3>
45                     <p>
46                         Creating attractive and responsive
47                         webpages using HTML5 and CSS3.
48                     </p>
```

contact.html (continued)

```
49         </article>
50         <article>
51             <h3>Frontend Development</h3>
52             <p>
53                 Developing user-friendly interfaces
54                 for web applications.
55             </p>
56         </article>
57         <article>
58             <h3>Responsive Design</h3>
59             <p>
60                 Designing webpages that work on
61                 different screen sizes.
62             </p>
63         </article>
64     </div>
65 </section>
66
67 <section id="contact">
68
69     <h2>Contact Us</h2>
70
71     <form>
72
73         <label for="name">Name:</label>
74         <input type="text" id="name" placeholder="Enter
75 your name">
76
77         <label for="email">Email:</label>
78         <input type="email" id="email" placeholder="Enter
79 your email">
80
81         <label for="message">Message:</label>
82         <textarea id="message" rows="5"
83             placeholder="Enter your message"></textarea>
84
85         <button type="submit">Submit</button>
86
87     </form>
88 </section>
89
90 </main>
91
92 <footer>
93     <p>&copy; 2026 HTML5 & CSS3 Webpage</p>
94 </footer>
95
96 </body>
97 </html>
```

7. CODE EXPLANATION

Code Element / CSS Feature	Explanation
<meta charset="UTF-8">	Sets the character encoding to UTF-8.
<meta name="viewport">	Sets the viewport configuration for responsive webpages.
<link rel="stylesheet">	References the CSS stylesheet used for styling the webpage.
<header>	Creates the top banner with the title and subtitle.
<nav> and <a>	Create navigation links to Home, About, Services and Contact.
<main>	Contains the main page content.
<section id="about">	Provides an About section explaining the roles of HTML5 and CSS3.
<section id="services">	Contains the services presented as three article elements.
<div class="container">	Groups the service articles for CSS Flexbox layout.
<article>	Represents each service card: Web Design, Frontend Development and Responsive Design.
<form>	Creates the contact form area.
<input type="email">	Provides an email input control.
<textarea>	Provides a multi-line message input area.
<button type="submit">	Provides the form submit button.
CSS rules	Style the header, navigation, sections, articles, form controls, button and footer.
@media (max-width: 700px)	Changes the service container to a column and displays navigation links as blocks on smaller screens.

8. TEST CASES AND EXECUTION DATA

The checks below cover the visible structure and controls implemented in contact.html and the CSS behavior represented by the Week 01 output.

Test	Test Case	Expected Result
TC-01	Page loading	The HTML5 & CSS3 webpage opens successfully.
TC-02	Navigation	Home, About, Services and Contact links are visible.
TC-03	Services layout	Three services are displayed in the services section.

Test	Test Case	Expected Result	Actual Result	Status
TC-04	Contact form	Name, email, message and submit controls are displayed.	All specified form controls are visible.	Pass
TC-05	Email field	The email field uses the email input type.	An email input control is rendered.	Pass
TC-06	Responsive rule	The stylesheet contains a max-width 700px media query.	The service container changes to column layout and navigation links become blocks at smaller widths.	Pass

9. OUTPUT

Output 1: HTML5 & CSS3 Webpage (reproduced from the repository output layout)

HTML5 & CSS3 Webpage

Learning Modern Web Development

[Home](#)
[About](#)
[Services](#)
[Contact](#)

About HTML5 & CSS3

HTML5 is used to create the structure of modern webpages, while CSS3 is used to design and style webpages.

Our Services

Web Design

Creating attractive and responsive webpages using HTML5 and CSS3

Frontend Development

Developing user-friendly interface for web applications.

Responsive Design

Designing webpages that work on different screen sizes.

Contact Us

Name:

Email:

Message:

Submit

The output shown above represents the webpage produced from the program source stored in the Week 01 repository.

10. RESULT

Thus, a webpage using HTML5 and CSS3 was successfully developed. The program uses semantic HTML5 structure, navigation, service articles, a contact form, CSS styling, Flexbox layout, form and button styling, hover effects, and a responsive media-query rule. The output was verified against the implemented webpage structure and the Week 01 repository screenshot.